

RAMCO INSTITUTE OF TECHNOLOGY
DEPARTMENT OF MECHANICAL ENGINEERING
ACADEMIC YEAR 2020-21
ME8811-PROJECT WORK

Sl.No.	Project Title	PO's Relevance	PSO's Relevance	Outcome
Projects in Identified Researches				
1.	Parametric Design and Investigation on the Effect of Machining Parameter in CNC Turning Machine.	1,2,3,4,5, 8,9,10,11,12	1,2	Applied for International Conference on Energy and Materials Technology organised by Sri Sivasubramaniya Nadar College of Engineering. The paper's review is in progress.
2.	Experimental Studies on Fused Filament Manufacturing of Polymer.	1,2,3,4, 8,9,10,11,12	2	The work has been completed and converted as a paper and to be converted as a paper and published in a Journal.
3.	Characterization and Study on Corrosion Behaviour of Copper Composition by Weight loss Method.	1,2,3,4,5, 8,9,10,11,12	1,2	The work has been completed and converted as a paper and to be converted as a paper and published in a Journal.
4.	Study on Corrosion Behaviour of Copper with Vanadium Reinforced Composites by Electrochemical Method.	1,2,3,4, 8,9,10,11,12	1,2	The paper is submitted for publication in Journal of Manufacturing Engineering which is published by Society of Manufacturing Engineers. The paper's review is in progress.
5.	Experimental Evaluation of Mixed Mode Fracture Toughness of Aluminium / Silicon Carbide Metal Matrix Composites.	1,2,3,4, 8,9,10,11,12	1,2	Presented the paper in International Virtual Conference on Emerging Trends in Engineering and Technology (ICETET2021) organised by ICCS College of Engineering and management, Thrissur, Kerala.
6.	Optimization and Simulation of Welding Butt Joint in Al 5083 by using MIG for Improving the Welding Process Parameters.	1,2,3,4,5, 8,9,10,11,12	1,2	The work has been completed and converted as a paper and to be converted as a paper and published in a Journal.
7.	Optimization of Process Parameters on DLP 3D Printed Coupons Using Taguchi Technique.	1,2,3,4,5,6,8,9,10,11,12	2	Paper presented in 3 rd national e-conference on Recent developments in Automobile and Mechanical Engineering (RDAME'21).

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8.	Analyzing Wear Behaviour of Ceramic Composite Coatings on Magnesium Alloy (AZ31) by using Friction Stir Processing.	1,2,3,4,5,8,9,10,11,12	2	The project is completed and it is planned to publish in a reputed Journal.
9.	Characterization of Stainless Steel - 316 Fabricated by WAAM Technique.	1,2,3,4,5,8,9,10,11,12	2	The project is completed and it is planned to publish in a reputed Journal
10.	Experimental Investigation on Piezo Enabled CFRP Composite Structure Under Static Loading Condition.	1,2,3,4,5,8,9,10,11,12	2	Paper Presented in NATIONAL WEB CONFERENCE ON CHALLENGES AND INNOVATION IN ENGINEERING AND TECHNOLOGY (NWCCIET-2021) 19th & 20th March 2021 Conducted by Ramco Institute of Technology, Rajapalayam.
11.	Experimental Investigation and Identifying the Suitable Process Parameters for Additively Manufactured PETG Material by Fused Deposition Modelling.	1,2,3,4,5,8,9,10,11,12	2	- Paper Presented in NATIONAL WEB CONFERENCE ON CHALLENGES AND INNOVATION IN ENGINEERING AND TECHNOLOGY (NWCCIET-2021) 19th & 20th March 2021 Conducted by Ramco Institute of Technology, Rajapalayam. - Mr. R. Venkatesh, AP/Mech Mr. S. Kathiravan, AP/Mech, Mr. R. Prabhakaran, AP/Mech, Mr. M. Ramar, AP/Mech, Mr. K. Amudhan, AP/Mech, Mr. J. Jerold John Britto, AP(SG)/Mech, Dr. S. Rajakarunakaran, Prof/Mech received the Best Paper Award in the International Conference on Energy and Materials Technology (ICEMT2021) organized by Department of Mechanical Engineering, SSN College of Engineering, Kalavakkam during 20-21 Aug., 2021.
12.	Deformation of Magnesium-Lithium based Composite during ECAP.	1,2,3,4,5,6,8,9,10,11,12	1,2	The project is completed and it is planned to publish in a reputed Journal.

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13.	Parametric design and investigation on the effect of Machining parameters and Form Tolerance in CNC Milling Machine.	1,2,3,4,5,6,8,9,10,11,12	2	Applied for International Conference on Energy and Materials Technology organised by Sri Sivasubramaniya Nadar College of Engineering. The paper's review is in progress.
14.	Experiment on CO ₂ reduction in metal active gas MAG welding and testing the mechanical properties of AISI 1025.	1,2,3,4,5,6,8,9,10,11,12	2	G. NithishKumar, C. Praveen, Mr. P. Pavithran, AP/Mech and Dr. S. Rajakarunakaran Prof & Head/ Mech, Presented a paper titled " EXPERIMENT ON CO ₂ REDUCTION IN METAL ACTIVE GAS (MAG) WELDING AND TESTING THE MECHANICAL PROPERTIES OF MILD STEEL AISI 1025 " in International Conference on Research & Developments in Engineering, Pharmacy & Management - 5th ICRDEPM-2021 organized by the International Institute of Research in Multidisciplinary – Skill Development Trust IIRM 2021, Andhra Pradesh on 06 June., 2021.
15.	Investigation and optimization of Additively manufactured thermoplastic material by fusion deposition modelling.	1,2,3,4,5,6,7, 8,9,10,11,12	1,2, 3,4	Presented the paper in NATIONAL WEB CONFERENCE ON CHALLENGES AND INNOVATION IN ENGINEERING AND TECHNOLOGY (NWCCIET-2021) 19th & 20th March 2021 Conducted by Ramco Institute of Technology, Rajapalayam.
16.	Experimental Performance comparison of specific moisture evaporation rate in heat pump dryer at fluidized bed, tray and compound methods for curry leaves.	1,2,3,4,5,6,7, 8,9,10,11,12	3	Presented the paper in the Second International Conference on Recent Challenges in Science, Engineering and Technology - 2021 (ICRCSET-2021 - Virtual conference) organized by Chalapathi Institute of Technology, Mothadaka, Andhra Pradesh.
17.	Finite Element Simulation on Carbon Reinforced Polymer Material under Uniaxial Tensile Loading using ANSYS ACP.	1,2,3,4,5,6,7, 8,9,10,11,12	1	Paper Presented in NATIONAL WEB CONFERENCE ON CHALLENGES AND INNOVATION IN ENGINEERING AND TECHNOLOGY (NWCCIET-2021) 19th & 20th March 2021 Conducted by Ramco Institute of Technology, Rajapalayam.

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18.	Effect of WAAM Process Parameters on Microstructure and Mechanical Properties of Stainless Steel 625.	1,2,3,4,5,6,7, 8,9,10,11,12	1,2,4	The project is completed and it is planned to publish in a reputed Journal.
19.	Experimental Performance comparison of Specific Moisture Evaporation Rate in Heat Pump dryer at fluidized bed, tray and compound methods for Bermuda Grass.	1,2,3,4,5,6,7, 8,9,10,11,12	1,2,3	Presented the paper in the Second International Conference on Recent Challenges in Science, Engineering and Technology - 2021 (ICRCSET-2021 - Virtual conference) organized by Chalapathi Institute of Technology, Mothadaka, Andhra Pradesh.
20.	Investigation and characterization of corrosion behaviour of copper composites by weight loss method	1,2,3,4,5,6,7, 8,9,10,11,12	2	The project is completed and it is planned to publish in a reputed Journal.
21.	Experimental Evaluation of Mixed Mode Fracture Toughness on Al/B4C Metal Matrix Composites.	1,2,3,4,5,6,7, 8,9,10,11,12	2	Presented the paper in the International Virtual Conference on “Emerging Trends in Engineering and Technology (ICETET 2021) organized by ICCS College of Engineering and Management, Mupliyam, Thrissur, Kerala, India on 2nd June 2021.
22.	Comparative Study between Virgin and Recycled material.	1,2,3,4,5,6,7, 8,9,10,11,12	2	The project is completed and it is planned to publish in a reputed Journal.
23.	Fabrication of Surface Ceramic Composites on Magnesium Alloy (AZ31) by using Friction Stir Processing with Direct Particle Injection Tool.	1,2,3,4,5,6,7, 8,9,10,11,12	2,4	The project is completed and it is planned to publish in a reputed Journal.
24.	Experimental Analysis of Single Basin Single Slope Solar Still with Chimney and Condensing Fan.	1,2,3,4,5,6,7, 8,9,10,11,12	1,2,3,4	The project is completed and it is planned to publish in a reputed Journal.
25.	Effect of Selective Laser Melting (SLM) Process Parameters on Microstructure and Mechanical Properties of 316L Stainless Steel.	1,2,3,4,5,6,7, 8,9,10,11,12	1,2,4	Mr. Ramar M, AP/Mech, presented a paper on the title “The Influence of Building Orientations On the Mechanical Characteristics of Selective Laser Melted SS316L” in ICEMT – 2021 organised by SSN College of Engineering.

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26.	Computational Simulation and Experimental Analysis of Free and Forced Fully Developed Air Flow in Green House Solar Dryer.	1,2,3,4,5,6,7, 8,9,10,11,12	3	The project is completed and it is planned to publish in a reputed Journal
27.	Design of Composite Laminate using Autodesk Heliux Composite and Non- Destructive Testing on CFRP Laminate.	1,2,3,4,5,6,7, 8,9,10,11,12	1	The project is completed and it is planned to publish in a reputed Journal.
28.	Experimental Investigation on the Effect of Reflecting Material of solar Parabolic Collector for Solar Water Heater.	1,2,3,4,5,6,7, 8,9,10,11,12	1,2,3	Presented the paper in International virtual conference conducted by ICCS college of Engineering and management, Thrissur, Kerala on 02.06.2021.
Projects in Specific Application				
29.	Building Smart Bot Design For Library Assistance.	1,2,3,4,5,6,7, 8,9,10,11,12	1	Registered for 21st International Conference on Hybrid Intelligent Systems and International Conference on Modelling, Simulation and Optimization 2021 hosted by NIT Silchar S. Valai Ganesh, V. Suresh, S. Godwin Barnabas, and S. Rajakarunakaran Published a book chapter titled “Building Smart Bot Design for Library Assistance”, in Robotics and Automation in Healthcare Advanced Applications, Apple Academy Press, and Hard ISBN: 9781774915486, 2023.
30.	Design and Analysis of 3D Printed Arm for Holding.	1,2,3,4,5,6, 8,9,10,11,12	1	The project is completed and it is planned to apply patent for the 3D printed arm.
31.	Experimental Investigation of Pappad Dryer Using Solar Parabolic Trough Collector.	1,2,3,4,5,6,7, 8,9,10,11,12	3	The project is completed and it is planned to apply patent for the Pappad dryer.
32.	Performance Study of Affordable Air Cooling System (AACS) for Tractor Cabinet.	1,2,3,4,5,6,7, 8,9,10,11,12	2,3	The project is completed and it is planned to apply patent for the of affordable air cooling system (AACS) for tractor cabinet.

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33.	Crack Detection Using Deep Learning.	1,2,3,4,5,6,7, 8,9,10,11,12	1,2,4	Presented the paper in 3rd International Conference on Mobile Radio Communications & 5G Networks, (MRCN-2021) 23 rd -24 th June, 2021 Organized by University Institute of Engineering & Technology, Kurukshetra University, Kurukshetra. The Conference paper will be published in Springer Journal after review.
34.	Design and Analysis of On Farm Cold Storage for Tomatoes.	1,2,3,4,5,6,7, 8,9,10,11,12	1,3	The project is completed and it is planned to apply patent for the On Farm Cold Storage for Tomatoes.
35.	Design and Optimization of Integrated Vertical Axis Wind Turbine.	1,2,3,4,5,6,7, 8,9,10,11,12	1,4	The project is completed and it is planned to publish it in a Reputed journal.
36.	Effective Oil Extrusion of Kappa 4C Oil Pump.	1,2,3,4,5,8,9,10,11,12	1	The project is completed and it is planned to publish it in a Reputed journal after performing additional work related to it.
37.	Comparative Study of Drying a Coconut Chip Using Conventional Method and Solar Dryer Using Sensible Storage Medium.	1,2,3,4,5,6,7, 8,9,10,11,12	1,3	The project is completed and it is planned to publish in a reputed Journal.
38.	Waste Water Treatment Using Crumb Rubber.	1,2,3,4,5,6,7, 8,9,10,11,12	1	The project is completed and it is planned to publish in a reputed Journal
39.	Parabolic Dish Collector With Auto Tracking System.	1,2,3,5,6,7, 8,9,10,11,12	3	The project is completed and it is planned to apply patent for the Parabolic Dish Collector.
40.	Experimentation Study of Spray Flash Water Distillation with LABVIEW Control system.	1,2,3,4,5,6,7, 8,9,10,11,12	1,2,3,4	The project was selected as one of the best projects on National Science Day Exhibition Conducted by Ramco Institute of Technology.
41.	Refrigerator with Adjustable Cooler Bag.	1,2,3,4,5,6,7, 8,9,10,11,12	1,2,3	The project is completed and it is planned to apply patent for the Cooler Bag.
42.	Automatic Speed Controller of Vehicle Near School and College Zones	1,2,3,4,5,6,7, 8, 9	1,2,4	The project is completed and it is planned to apply patent for the Automatic speed controller

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43.	Removal of Moisture in Paper Production	1,2,3,4,5,6,7, 8,9,10,11,12	1,2,3,4	Presented a paper in NATIONAL WEB CONFERENCE ON CHALLENGES AND INNOVATION IN ENGINEERING AND TECHNOLOGY (NWCCIET-2021) 19th & 20th March 2021 Conducted by Ramco Institute of Technology, Rajapalayam
44.	Error Analysis of Robotic Arm.	1,2,3,4,5,6,7, 8,9,10,11,12	1,2	Mr. Valai Ganesh S and Selva Sundar T presented a paper on "Error Analysis in Robotic ARM", in National Web Conference on Challenges and Innovation in Engineering and Technology – 2021, Organized by RIT
45.	Studies on Dynamic Behaviour of Control Arm in Steering System.	1,2,3,4,5,6,7, 8,9,10,11,12	1,2	Presented a paper in International Virtual Conference on Emerging Trends in Engineering and Technology – ICETET- 2021 held at ICCS College of Engineering and Management (ICCSCM),Mupliyam, Thrissur, Kerala, India on May 19, 2021.
46.	Conversion of Waste Rubber Tyre into Composite Material.	1,2,3,4,5,6,7, 8,9,10,11,12	1,2	Presented the paper in NATIONAL WEB CONFERENCE ON CHALLENGES AND INNOVATION IN ENGINEERING AND TECHNOLOGY (NWCCIET-2021) 19th & 20th March 2021 Conducted by Ramco Institute of Technology, Rajapalayam.
Solution to Industrial Problems based Projects				
47.	Automation in Cement Bag Feeding in Electronic Rolling Packer.	1,2,3,4,5,6,8,9,10,11,12	1,2	Presented the paper in National level symposium on "Recent Trends in VLSI Signal Processing, IoT, Communication and Embedded Systems- VSPICE 2021, Conducted by dept of ECE of Amirtha College of Engineering and Technology, Nagercoil.
48.	Automation of Reject Handling at Truck Loading area.	1,2,3,4,5,6,8,9,10,11,12	1	Hari Prasath M did this project in Ramco Cements and the company has implemented the project.
49.	Design and Impact Test on Car Door Using Sandwich panel.	1,2,3,4,5,6,8,9,10,11,12	1,4	The project is completed and it is planned to apply patent for the design.

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50.	Design and Prediction of Three Dimensional Temperature Distribution of Solid Body Using Machine Learning.	1,2,3,4,5,6,7, 8,9,10,11,12	1,2,3,4	A paper was published by S. Maharajan as a first author on the title “Prediction of Temperature Distribution in Three Dimensional Solid Objects using COMSOL and Python” in Materials Today: Proceedings, Volume 66, Part 3, 2022, Pages1241-1246. https://doi.org/10.1016/j.matpr.2022.05.062 .
51.	Implementation of KAIZEN Concept in Textile Industry.	1,2,3,4,5,6,7,8,9,10,11,12	2	Presented a paper in NATIONAL WEB CONFERENCE ON CHALLENGES AND INNOVATION IN ENGINEERING AND TECHNOLOGY (NWCCIET-2021) 19th & 20th March 2021 Conducted by Ramco Institute of Technology, Rajapalayam
52.	Design and Development of CFD Model to Measure Variable Flow Rate.	1,2,3,4,5,6,7, 8, 10,11,12	1,2,3,4	The project was selected as one of the best projects on National Science Day Exhibition Conducted by Ramco Institute of Technology on 28.02.2020
53.	Maintaining the Quality of Liquid Medicine by Controlling its Temperature.	1,2,3,4,5,6,7, 8,9,10,11,12	3	Rijith Kumar V and Rohith S did this project in Aravind Herbals Private Limited and their project work was appreciated by the company.